14/10/15 Chapter 5: Special Distributions

Discrete: Bernoulli(p)

MGF: $1-p+pe^t$

Binomial(n, p)

MGF: $(1-p+pe^t)^n$

$X_i \sim \text{Binomial}(n_i, p)$ indep

$i = 1, \dots, k$

$Y = X_1 + X_2 + \cdots + X_k$

$Y \sim \text{Binomial}(n_1 + \cdots + n_k, p)$

$M_Y(t) = \prod_{i=1}^{k} M_{X_i}(t)$

$(1-p+pe^t)^{n_i}$

$= (1-p+pe^t)^{n_1+\cdots+n_k}$

Geometric(p)

# trials till first success	# failures till first success	Pr
1	0	p
2	1	$(1-p)p$
3	2	⋮
⋮	⋮	
k	$k-1$	$(1-p)^{k-1}p$
⋮	⋮	⋮

$E[X]$: $1/p$ | $1/p - 1$

$\text{Var}(X)$: $\frac{1}{p^2} - \frac{1}{p}$? | $\frac{1}{p^2} - \frac{1}{p}$?

MGF: $\sum_{k=1}^{\infty} (1-p)^{k-1} p e^{kt} = \frac{pe^t}{1-(1-p)e^t}$ | $\frac{pe^{t-1}}{1-(1-p)e^t}$ | $(1-p)e^t < 1$

Negative binomial (r, p)

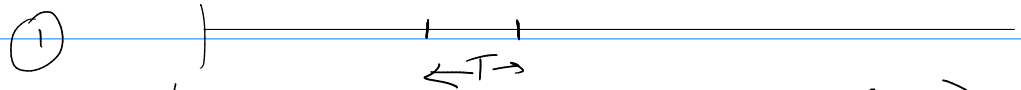
# trials till r-th success	# failures till r-th success	Pr
r	0	
$r+1$	1	
⋮	⋮	
$k+r$	k	$\binom{k+r-1}{r-1} p^r (1-p)^k$

$X_1 + X_2 + \cdots + X_r \sim$ Negative binomial (r, p)

$X_i \sim$ Geometric (p) indep

Poisson (λ)

Poisson process rate: α per time unit

① 

arrivals $\sim$ Poisson (αT)

② Arrivals in non-overlapping time windows are independent

Binomial $(n, p) \approx$ Poisson (λ)

$np = \lambda$

$p \to 0$

$n \to \infty$

$X \sim$ Poisson (λ) $p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$ $k = 0, 1, 2, \cdots$

$E[X] = \lambda$ $\mathrm{Var}(X) = \lambda$ $M_X(t) = e^{\lambda(e^t - 1)}$

$X_i \sim \text{Poisson}(\lambda_i)$, indep

$i = 1, 2, \ldots, n$

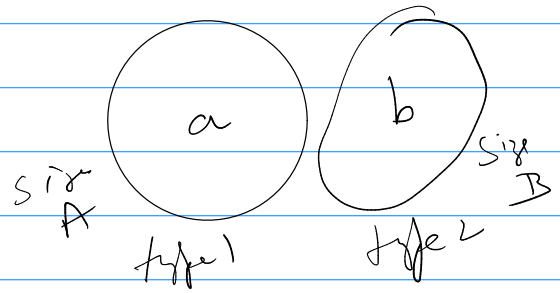
$$X_1 + X_2 + \cdots + X_n \sim \text{Poisson}(\lambda_1 + \cdots + \lambda_n)$$

Hypergeometric distribution

→ Sampling without replacement

A, B: two integers

$A+B$: total population



Select n at random without replacement

$n = 0, \ldots, A+B$

Samples



Select A positions in one of $\binom{A+B}{A}$ ways.

Place a's in selected positions

Place b's in other positions.

X = # a's in sample $\in \{\max(0, n-B), \ldots, \min(n, A)\}$

$$\Pr(X=k) = \frac{\binom{A}{k}\binom{B}{n-k}}{\binom{A+B}{n}}$$

$X = X_1 + X_2$

$X = I(\text{a in pos. } 1) + I(\text{a in pos. } 2) + \cdots + I(\text{a in pos } n)$

$$E[X] = \frac{nA}{A+B} \qquad \text{Var}(X) = n\frac{A}{A+B}\cdot\frac{B}{A+B} + n(n-1)\text{Cov}(X_1, X_2)$$

Reference: A, B: very large $\quad p = \frac{A}{A+B}$ $\quad$ Hypergeometric$(n, A, B) \simeq$ Binomial(n, p)

Bernoulli(p), Binomial(r, p), Poisson(λ),
Geometric(p), Neg-Binom(r, p), Hypergeometric(n, A, B)